Problem Set 2

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Problem 1

a)

The separation rate s is constant. Labor productivity z follows a discrete Markov chain with transition matrix $\pi(z' \mid z)$. Let $\beta = 1/(1+r)$ denote the discount factor and $\gamma \in (0,1)$ workers' bargaining power.

Value Functions

Unemployed worker:

$$U_z = b + \beta\, \mathbb{E}_z\big[f(\theta_{z'})\, E_{z'} + \big(1 - f(\theta_{z'})\big)\, U_{z'}\big] \tag{1}$$

Employed worker:

$$E_z = w_z + \beta\, \mathbb{E}_z[(1-s)\, E_{z'} + s\, U_{z'}] \tag{2}$$

Filled job:

$$J_z = z - w_z + \beta(1-s)\, \mathbb{E}_z[J_{z'}] \tag{3}$$

Equilibrium Conditions

Free entry:

$$c = q(\theta_z)\, J_z \tag{FE}$$

Nash bargaining over surplus $S_z \equiv J_z + E_z - U_z$:

$$E_z - U_z = \gamma S_z, \qquad J_z = (1-\gamma)\, S_z \tag{NB}$$

Equilibrium Equation

Define the match surplus S_z. Subtracting the unemployed worker's equation from the employed worker's, adding the firm's equation, and applying (NB) to eliminate wages yields the surplus equation. Using (FE) to substitute $J_z = c/q(\theta_z)$, so that:

$$S_z = \frac{c}{(1-\gamma)\, q(\theta_z)}, \tag{4}$$

and dividing through by $c/(1-\gamma)$ and using $f(\theta) = \theta\, q(\theta)$, the surplus equation reduces to a single equation that implicitly defines θ_z for each state z:

$$\boxed{\frac{1}{q(\theta_z)} = (1-\gamma)\frac{z-b}{c} + \beta(1-s)\,\mathbb{E}_z\left[\frac{1}{q(\theta_{z'})}\right] - \beta\gamma\,\mathbb{E}_z[\theta_{z'}]} \tag{5}$$

The three terms have a clean interpretation: the expected cost of filling a vacancy equals the net flow surplus from the match, plus the discounted continuation value of the filled job, minus the share of future market tightness that accrues to the worker through bargaining.

Wage Equation

Applying (NB) to the value of unemployment and rearranging:

$$w_z = (1-\gamma)b + \gamma z + \beta\gamma c\,\mathbb{E}_z[\theta_{z'}] \tag{6}$$

Wages absorb a fraction γ of both the flow surplus and expected future market tightness. This is the source of the amplification failure: when z rises, $\mathbb{E}_z[\theta']$ rises too, so wages increase by nearly as much as productivity, leaving firms' profits—and hence vacancy creation incentives—largely unchanged.

Unemployment Dynamics

$$u_{t+1} = s(1-u_t) + \big(1 - f(\theta_{z_t})\big)\, u_t \tag{7}$$

with steady state:

$$u^* = \frac{s}{s + f(\theta_z)} \tag{8}$$

b)

We discretize the AR(1) process following Rouwenhorst (1995). Details are explained in the code.

c)

We calibrate the model following Table 2 in Shimer (2005) , but adjusting x, y and z to be monthly. Table 1 shows the calibrated parameters.

d)

Table 2 reports the results of our replication of Table 3 in Shimer (2005). Overall, the model reproduces the main patterns in the paper quite closely. Unemployment is strongly negatively correlated with vacancies, while vacancies, market tightness $\frac{v}{u}$, the job-finding rate f, and productivity z are all strongly positively correlated.

Several correlations are very close to one in absolute value. This occurs because productivity is the only aggregate shock in the model, so most endogenous variables move closely together with the same underlying state.

The remaining moments are also close to those reported in Shimer (2005). The main small differences appear in the autocorrelations of $\frac{v}{u}$ and f, which may arise from the monthly calibration, the discretization of the productivity process, and simulation error. Overall, the replication matches the qualitative results in the original paper.

Despite matching the qualitative correlations, the model still generates too little volatility in labor market variables relative to the data, which is the essence of the Shimer puzzle.

Problem 2

a)

Table (3) shows the results of replicating Shimer (2005) but with a higher value for leisure. Following Hagedorn and Manovskii (2008), we increase the value of leisure z to be close to average productivity, so that the average surplus from a match $(p^* - z)$ is small. In their original calibration, Hagedorn and Manovskii set $b \approx 0.955$, which implies that workers are nearly indifferent between employment and unemployment. However, implementing this value directly in our discrete-time framework is not feasible: the Rouwenhorst (1995) discretization with $N = 21$ grid points generates a minimum productivity value of $p_{min} \approx 0.663$, and any value of z exceeding p_{min} would violate the positive-surplus condition $p_i - z > 0$ required for matches to be mutually beneficial at all points of the state space. We therefore set $z = 0.95 \times p_{min} \approx 0.630$, which is the highest feasible value of leisure consistent with positive surplus everywhere on the grid.

b)

Table (4) shows the results of the replication of Shimer (2005) , while keeping wages constant after a match is realized -following the idea of Hall (2005).

c)

Table (5) follows the Chodorow-Reich and Karabarbounis (2016) critique and shows the results with a procyclical b. In their paper, they propose setting $z(p) = \zeta p$, where $\zeta \in (0, 1)$ is a scalar that determines both the average level of the outside option and how strongly it responds to aggregate productivity. When productivity p is high, workers enjoy better outside opportunities, and when it is low, the outside option deteriorates proportionally. The parameter ζ is calibrated so that the procyclical leisure specification is consistent with the Hagedorn and Manovskii (2008) calibration at the steady state. Specifically, evaluating at the mean productivity $p^* = 1$ gives $z(p^*) = \zeta \times 1 = 0.955$, which exactly matches the value of leisure that Hagedorn and Manovskii target in their original calibration. Note that by construction this is a higher value of b than the one in part (a), where we followed Shimer (2005) and set $z = 0.40$.

d)

Table 6 reports the results when the discount factor is allowed to vary with productivity according to

$$\beta(z) = \bar{\beta} + \phi(z - \bar{z}),$$

where $\bar{\beta}$ is the baseline monthly discount factor, $\bar{z} = 1$ denotes mean productivity, and ϕ determines the sensitivity of discounting to the state of the economy. In periods of high productivity, agents discount future payoffs slightly less, which increases the continuation value of a match and may strengthen firms' incentives to post vacancies.

However, under our calibration the variation in $\beta(z)$ is quantitatively small, so the effect on equilibrium outcomes is limited. As shown in Table 6, the volatility of unemployment, vacancies, and labor market tightness remains very close to the baseline case. Consequently, allowing the discount factor to be procyclical does not substantially amplify productivity shocks, and the model remains close to the baseline specification.

e)

Comparing Tables 2 and 3, the Hagedorn–Manovskii calibration generates higher volatility of unemployment u, vacancies v, and market tightness v/u than the baseline model. The intuition is that increasing the value of leisure b reduces the match surplus $z - b$. When this surplus is small, even small productivity shocks produce large proportional changes in firms'

profits, which makes vacancy creation much more sensitive to fluctuations in productivity. As a result, labor market variables become more volatile than in the baseline model.

Comparing Tables 2 and 4, the results change dramatically. This is normal. Under Hall's constant-wage assumption, wages no longer adjust to productivity shocks. In the baseline model, Nash bargaining makes wages move closely with productivity, so most of the shock is absorbed by wages and firms' profits change little. When wages are fixed, productivity shocks translate directly into changes in firms' profits. This makes vacancy creation much more responsive, generating much higher volatility in vacancies, unemployment, and market tightness. Hence, wage rigidity is one possible way to address the Shimer puzzle.

Comparing Tables 2 and 5, the results are very similar to the baseline, and in some dimensions volatility is even lower. This is consistent with the critique of Chodorow-Reich and Karabarbounis (2016). If leisure is procyclical, so that $b(z) = \zeta z$, then the worker's outside option increases in good times. As a consequence, the match surplus $z - b(z)$ changes little over the cycle. Firms' profits therefore fluctuate less, vacancy creation reacts weakly, and the model continues to generate low labor-market volatility.

Finally, comparing Tables 2 and 6, the results remain very close to those of the baseline model. Introducing a procyclical discount factor implies that agents value future match surplus slightly more in periods of high productivity. In principle, this could increase the incentive to create vacancies, since firms place a higher value on future profits. However, in our calibration the variation in the discount factor is small, so this mechanism has only a limited effect on equilibrium outcomes. As a result, the volatility of unemployment, vacancies, and labor market tightness remains very similar to the baseline case, and the model still generates only modest amplification of productivity shocks.

Appendix

References

Chodorow-Reich, Gabriel and Loukas Karabarbounis, "The Cyclicality of the Opportunity Cost of Employment," *Journal of Political Economy*, 2016, *124* (6), 1563–1618.

Hagedorn, Marcus and Iourii Manovskii, "The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited," *American Economic Review*, 2008, *98* (4), 1692–1706.

Hall, Robert E., "Employment Fluctuations with Equilibrium Wage Stickiness," *American Economic Review*, 2005, *95* (1), 50–65.

Shimer, Robert, "The Cyclical Behavior of Equilibrium Unemployment and Vacancies," *American Economic Review*, March 2005, *95* (1), 25–49.

Table 1: Parameter Values

Parameter	Symbol	Value	Source / Target
Panel A. Frequency-invariant parameters			
Mean productivity	$\bar{z}$	1.00	Normalization
Value of leisure	b	0.40	Shimer (2005)
Matching elasticity	α	0.72	Estimated
Bargaining power	γ	0.72	Hosios: $\gamma = \alpha$
Panel B. Monthly flow parameters			
Discount factor	β	0.99602	$(1 + r_q)^{-1/3}$, $r_q = 0.012$
Separation rate	s	0.034	Shimer (2005), Table 1
Matching function constant	μ	0.450	$f(\theta^*) = 0.45$ at $\theta^* = 1$
Vacancy posting cost	c	0.210	Free-entry at $\theta^* = 1$
Panel C. Productivity process (monthly AR(1))			
Autocorrelation	ρ	0.99867	$e^{-\kappa/3}$, $\kappa = 0.004$
Innovation std. dev.	σ_ε	0.00952	$\sigma_q/\sqrt{3}$, $\sigma_q = 0.0165$
Unconditional std. dev.	σ_z	0.1845	$\sigma_\varepsilon/\sqrt{1-\rho^2}$
Panel D. Discretization and filtering			
Grid points	N	21	Rouwenhorst (1995)
HP filter parameter	λ	10^5	Hodrick–Prescott, quarterly

Notes: Monthly parameters are converted from Shimer (2005)'s quarterly calibration. The matching function is $f(\theta) = \mu\theta^{1-\alpha}$ with $q(\theta) = \mu\theta^{-\alpha}$. Vacancy cost c is solved from the steady-state free-entry condition $c = (1-\gamma)(\bar{z}-b)/\big[\mu^{-1}(1-\beta(1-s)) + \beta\gamma\big]$. The productivity process $z_{t+1} = \rho\, z_t + \varepsilon_{t+1}$ is discretized using Rouwenhorst's (1995) method, which exactly matches the conditional and unconditional first two moments for any N.

Table 2: Baseline (Shimer 2005)

	u	v	v/u	f	z
Standard deviation	0.009	0.026	0.034	0.009	0.020
	(0.002)	(0.006)	(0.008)	(0.002)	(0.005)
Quarterly autocorrelation	0.936	0.883	0.912	0.912	0.912
	(0.020)	(0.038)	(0.028)	(0.028)	(0.028)
Correlation matrix					
u	1.000	-0.937	-0.964	-0.964	-0.964
	(0.000)	(0.033)	(0.025)	(0.025)	(0.025)
v		1.000	0.996	0.996	0.995
		(0.000)	(0.010)	(0.010)	(0.010)
v/u			1.000	1.000	1.000
			(0.000)	(0.000)	(0.001)
f				1.000	1.000
				(0.000)	(0.001)
z					1.000
					(0.000)

Table 3: High Leisure (Hagedorn–Manovskii 2008)

	u	v	v/u	f	z
Standard deviation	0.014	0.043	0.056	0.016	0.020
	(0.004)	(0.011)	(0.015)	(0.004)	(0.005)
Quarterly autocorrelation	0.936	0.883	0.911	0.912	0.912
	(0.020)	(0.038)	(0.028)	(0.028)	(0.028)
Correlation matrix					
u	1.000	-0.936	-0.964	-0.964	-0.961
	(0.000)	(0.034)	(0.025)	(0.025)	(0.025)
v		1.000	0.996	0.996	0.992
		(0.000)	(0.010)	(0.010)	(0.011)
v/u			1.000	1.000	0.997
			(0.000)	(0.000)	(0.004)
f				1.000	0.997
				(0.000)	(0.004)
z					1.000
					(0.000)

Table 4: Constant Wage (Hall 2005)

	u	v	v/u	f	z
Standard deviation	0.294	3.777	4.025	0.790	0.020
	(0.233)	(3.431)	(3.624)	(0.697)	(0.005)
Quarterly autocorrelation	0.947	0.878	0.891	0.900	0.912
	(0.045)	(0.101)	(0.094)	(0.086)	(0.028)
Correlation matrix					
u	1.000	-0.617	-0.710	-0.727	-0.617
	(0.000)	(0.627)	(0.471)	(0.460)	(0.317)
v		1.000	0.924	0.914	0.590
		(0.000)	(0.251)	(0.287)	(0.332)
v/u			1.000	0.990	0.598
			(0.000)	(0.021)	(0.324)
f				1.000	0.609
				(0.000)	(0.322)
z					1.000
					(0.000)

Table 5: Procyclical Leisure (Chodorow-Reich & Karabarbounis 2016)

	u	v	v/u	f	z
Standard deviation	0.005	0.015	0.020	0.006	0.020
	(0.001)	(0.004)	(0.005)	(0.001)	(0.005)
Quarterly autocorrelation	0.936	0.883	0.912	0.912	0.912
	(0.020)	(0.038)	(0.028)	(0.028)	(0.028)
Correlation matrix					
u	1.000	-0.937	-0.964	-0.964	-0.964
	(0.000)	(0.033)	(0.025)	(0.025)	(0.025)
v		1.000	0.996	0.996	0.996
		(0.000)	(0.010)	(0.010)	(0.010)
v/u			1.000	1.000	1.000
			(0.000)	(0.000)	(0.000)
f				1.000	1.000
				(0.000)	(0.000)
z					1.000
					(0.000)

Table 6: Procyclical Discount Factor

	u	v	v/u	f	z
Standard deviation	0.009	0.026	0.034	0.010	0.020
	(0.002)	(0.006)	(0.008)	(0.002)	(0.005)
Quarterly autocorrelation	0.936	0.883	0.912	0.912	0.912
	(0.020)	(0.038)	(0.028)	(0.028)	(0.028)
Correlation matrix					
u	1.000	-0.937	-0.964	-0.964	-0.964
	(0.000)	(0.033)	(0.025)	(0.024)	(0.025)
v		1.000	0.996	0.996	0.995
		(0.000)	(0.010)	(0.010)	(0.010)
v/u			1.000	1.000	1.000
			(0.000)	(0.000)	(0.001)
f				1.000	1.000
				(0.000)	(0.001)
z					1.000
					(0.000)